Benchmarking 1D-RS, 2D-RS, and Block-Circulant codes

A controlled comparison of erasure-coding schemes as candidates for Ethereum's data-availability sampling: one-dimensional Reed-Solomon (what PeerDAS uses today), two-dimensional Reed-Solomon, and Block-Circulant (BC) codes at five parameterisations. All of them run over the same field, on the same blob, under the same erasure model, and are judged by the same success criterion.

The question each scheme has to answer is the same: **if a fraction of the encoded data goes missing, can the original blob still be reconstructed exactly — and what did that cost in storage and in decode time?**

This revision changes two things, both following supervisor guidance of 2026-09-15 and both aimed at making the comparison fair rather than at making any scheme look better.

2D-RS moves from an 88×88 grid to 88×96, putting its rate at 0.485 against the others' 0.500 — the closest any grid reachable in this field gets to 0.5, and a correction of the previous 0.529. The natural-looking 90×90 turns out to be unbuildable here, for a reason worth stating up front because it is easy to miss: a length-n NTT needs n to divide $p-1$, and $90 = 2 \cdot 3^2 \cdot 5$ needs both a 3^2 (this field's $p-1$ carries only 3^1) and a 5 (absent entirely).

The BC rows move from $\lambda = \mu$ to $\lambda \in \{2, 3\}$, with one $\lambda = 4$ row kept as an anchor. The previous set swept $\lambda = \mu$ at $\mu = 4, 8, 16$, which measured the same extreme three times and pushed the decoder's reach down to 0.059. $\lambda = 16$ is not implementable, and the source paper's algorithms assume $\lambda = 2$, so the sweep now sits where the construction actually lives. This turns out to matter a great deal: the best BC row now tolerates 0.35 rather than 0.20.

The practical effect of both changes is that every scheme in the table below stores within 3% of twice its blob, so differences in what they tolerate are differences between the codes and their decoders, not between their budgets.

How the code works

The whole experiment is five moving parts. Each is a separate file, and the benchmark drives them without reimplementing any of their logic.

Packing a blob (`blob_generator.py`). An Ethereum blob is 4096 field elements of the BLS12-381 scalar field. Each element is 32 bytes on the wire but only 31 are usable, because the value must stay below the field modulus — so one blob carries 126,976 bytes of application data. This is the unit everything else operates on: a list of 4096 field elements, and a function to turn it back into the original bytes.

1D-RS (`1D_RS_Galois.py`). The key idea is that the blob's 4096 elements are not raw data to be copied — they are the *values of a single polynomial* sampled at 4096 fixed points. Encoding recovers that polynomial (an inverse NTT turns 4096 values into 4096 coefficients), pads it with zeros, and evaluates it again at 8192 points (a forward NTT). The result is 8192 symbols describing the same polynomial. Since a polynomial of degree below 4096 is pinned down by *any* 4096 of its values, any 4096 surviving symbols reconstruct everything. Decoding builds an erasure-locator polynomial whose roots sit exactly at the missing positions, multiplies through by it, inverts the transform, and divides the locator back out —

recovering the original polynomial in one pass. Crucially it first checks whether the number of erasures exceeds `n−k`; if so it gives up immediately rather than attempting anything.

2D-RS (`2D_RS_Galois.py`). The same interpolate-and-extend step, applied twice. The blob is reshaped into a 64×64 grid; every row is extended, then every column of that result is extended. Because the original grid is a single bivariate polynomial, every row *and* every column of the result is independently a valid 1D-RS codeword. Decoding exploits this: it sweeps all columns, then all rows, repeatedly. A line holding too many erasures to fix alone often becomes fixable once the perpendicular sweep has repaired some of its cells, so repairs cascade across the two axes until nothing more can be done.

What changed here is the extension factor, twice. A product code's rate is the *product* of its two component rates, so doubling both axes — the original 64→128 setting — gives rate 1/4 and 4× storage. That made the first comparison unfair: 2D-RS was being judged at the same erasure rate as codes spending half as much. The fix is to pick the component rate so that its square lands on 0.5: each axis should extend $64 \to 64/\sqrt{0.5} = 90.5$, which is where the "$64 \times \sqrt{2} \approx 90$, so use 90×90" suggestion comes from. The reasoning is right; the number is unavailable.

A length-n NTT exists only if n divides $p-1$, and for the BLS12-381 scalar field

> $p - 1 = 2^{32} \cdot 3 \cdot 11 \cdot 19 \cdot 10177 \cdot 125527 \cdot 859267 \cdot 906349^2 \cdot 2508409 \cdot 2529403 \cdot \ldots$

$90 = 2 \cdot 3^2 \cdot 5$ needs a 3^2 where only 3^1 is available, and a 5 that does not appear at all. There is no length-90 NTT here, so no 90×90 grid. The reachable lengths above 64 are 66, 76, 88, 96, 114, 128, 132, … — and 88 and 96 bracket the 90.5 target from either side, with geometric mean $\sqrt{8448} = 91.9$ sitting close to it. Component codes need not be equal, so the grid is now **88 × 96**: component rates 64/96 = 0.667 and 64/88 = 0.727, overall 4096/8448 = **0.485** at 2.06× storage. That is the closest to 0.5 of any reachable pair. The previous 88×88 gave 0.529 at 1.89×, further off on both counts *and* cheaper than the codes it was compared against.

Exactly 0.5 is unreachable regardless: it would need the codeword to be 8192 cells, a power of two forces both grid dimensions to be powers of two, and each axis must at least double to carry any parity at all — which lands on 16384, not 8192.

Block-Circulant (`BCcodeEthereum.py`). Structurally different: instead of one global polynomial, the message is cut into μ small blocks of ω symbols, and each block gets ρ parity symbols computed from its own information plus that of λ consecutive neighbours, wrapped around a ring. Overhead is 2×, the same as 1D-RS.

The parameterisation matters more than it first appears, and λ turns out to be the parameter that matters. Two quantities move in *opposite* directions as it changes:

- The **code's** guarantee. Minimum distance is $d = \lambda\rho + 1$, so the fraction of erasures it survives against *any* pattern is $(d-1)/n = \lambda\rho/n$. This **rises** with λ.
- The **decoder's** reach. `DecCBC[2]` resolves a local block when the erasures landing in its support of $N_O = \lambda\omega + \rho$ symbols number at most ρ, i.e. while the erasure rate is under $\rho/N_O = 1/(1+\lambda)$ when $\rho = \omega$. This **falls** with λ.

So a larger λ buys a better code and a worse decoder. $\lambda = \mu$ is the extreme: d reaches $4097 = n - k + 1$, the Singleton bound, making the code MDS — exactly as good as Reed-Solomon — while the decoder's reach bottoms out at $1/(1+\mu)$.

That is what the previous revision swept, at μ = 4, 8, 16: three MDS codes whose decoders reached 0.200, 0.111 and 0.059. It measured the same extreme three times and never sampled the regime the construction is for. Supervisor guidance is that `DecCBC` is practical only to λ = 3, that λ = 16 is not implementable, and that analysis belongs at λ = 2 and 3 — which is also where the source paper sits, as the next section shows. The rows swept here follow that, keeping one λ = 4 row as the MDS anchor.

Note what that produces: **ω2048 λ2 and ω1024 λ4 are both MDS with $d = 4097$ — identical as codes — yet their decoders reach 0.34 and 0.20.** Same code quality, different reach, from λ alone. Separating the code from the decoder is what this sweep measures, and that pair is the cleanest statement of it.

Checking the parameters against the source

The BC construction comes from Sasidharan, Viterbo and Feist, *Block Circulant Codes for Ethereum PeerDAS* (ePrint 2026/708). Its notation table matches this implementation exactly — [No, Ko] = [λω + ρ, λω], [n, k] = [μ(ω+ρ), μω], $R = \omega/(\omega+\rho)$, $S = \lambda\rho/n$, So = ρ/No — which is worth recording, because two of those are the quantities this report now reports and one of them was derived here independently before the paper was consulted:

- The paper's **stopping rate** $S = \lambda\rho/n$ is precisely the $(d-1)/n$ erasure tolerance used throughout below.
- The paper's **local stopping rate** So = ρ/No is precisely the decoder reach predicted above. The prediction and the paper's definition are the same quantity, arrived at separately.

Two further points from it bear on the rows chosen here. First, the paper **fixes λ = 2** for its encoding and reconstruction algorithms and for its entire performance analysis, and states the λρ erasure-correction result for **λ = 2, 3 only**. The λ = 2 rows therefore sit exactly where the paper does, the λ = 3 row sits at the edge of its stated range, and the λ = 4 row is outside it — its $d = 4097$ extrapolates a formula past where it was proved, which is another reason to read it as an anchor rather than a result.

Second, the paper's worked example is μ = 4, λ = 2, ρ = ω with **ω = 32 cells at stacking depth $D = 64$** — it counts ω and ρ in *cells*, not in field symbols. Converting: a 4096-symbol blob at $D = 64$ is 64 cells, so μ = 4 gives ω = 16 cells = **1024 field symbols**. The ω1024 λ2 row below *is* the paper's worked configuration, expressed in symbols.

The harness (`benchmark_1D_2D_BC.py`). Each scheme is wrapped in a small adapter exposing the same five operations — `encode`, `random_mask`, `corrupt`, `decode`, `extract_message` — so the sweep loop never needs to know that 2D-RS's codeword is a grid while BC's is a flat vector. The adapters call the real functions in the files above; no coding logic is duplicated, so fixing one of those files changes the benchmark's results automatically.

The experimental pipeline

Every measurement follows the same seven steps, and only two of them differ between schemes:

1. **Generate** 126,976 random application bytes.
2. **Pack** them into one blob of 4096 field elements, the real EIP-4844 way.
3. **Encode** — *scheme-specific* — expanding 4096 elements into a codeword of `n` elements.
4. **Erase** — withhold each codeword position independently with probability equal to the target rate, simulating samples that peers failed to provide.
5. **Decode** — *scheme-specific* — attempt to reconstruct the full codeword from what survived.

6. **Extract** the original 4096 elements back out of the recovered codeword.
7. **Verify** by unpacking to bytes and comparing against step 1.

Steps 1, 2, 4, 6 and 7 are byte-identical across every scheme. That is what makes the comparison fair: any difference in the results has to come from step 3 or step 5, which is precisely what is being compared.

Step 7 is the one worth dwelling on. Success is defined as *the recovered bytes equal the original bytes* — not as "the decoder reported success", and not even as "the field elements matched". That distinction is not pedantry: one of these decoders turned out to report success on data it had silently corrupted, and only the byte comparison caught it.

What is held identical

- **Field** — the real BLS12-381 scalar field.
- **Message** — the same 4096 elements, packed from the same bytes, fed to every encoder. Not one random blob per scheme.
- **Erasure model** — independent per-position withholding at a shared target rate. 1D-RS and all five BC rows share `n = 8192`, so on each trial they are corrupted at the *same literal positions*, not merely the same rate. 2D-RS has `n = 8448` and draws its own masks at the same rate.
- **Success criterion** — the full byte-level round trip described above.

Storage is matched to within 3%: 2.00× for 1D-RS and every BC row, 2.06× for 2D-RS. The small residual difference is reported rather than controlled away, and it runs *against* 2D-RS rather than for it.

Method

Erasure rates are swept from 2% to 55%. Trial counts vary by scheme and rate: a scheme decoding against a formal budget check behaves deterministically well away from its threshold, so one trial there is representative, while near a threshold the outcome genuinely varies from draw to draw and needs several. Trials are therefore concentrated around each scheme's own transition — which is not the same rate for all of them. Each success rate carries a 95% Wilson confidence interval.

How thresholds are estimated

A scheme's **threshold** is the erasure probability at which it recovers the blob half the time. The erasure probability is the per-symbol parameter ε of a binary erasure channel BEC(ε) — the target rate the sweep feeds to its random mask — which is the standard axis for this in coding theory. The fraction of symbols actually erased in a given trial is a random variable around ε, not a better axis: a threshold is a property of the channel a code can survive.

The sweep samples each threshold densely. It covers 2% to 55% erasure to show every scheme's whole curve, and merges in a fine grid — steps of 0.005, or 0.01 where a transition is gentler — across each scheme's own transition band, with 12–20 trials per rate inside that band and one or two outside it, where the outcome is certain either way. Because there is one grid for every scheme, the n = 8192 schemes are corrupted at literally the same positions at every rate, thresholds included.

An earlier version of this report spread its trials evenly and ran as few as three near a threshold. That is enough to show a curve's shape but not to place a threshold to two decimal places, and it showed:

- **1D-RS came out at 0.505**, above the 0.5 that an MDS code with n = 2k can sustain. The decoder was not at fault — of the three trials at ε = 0.50, the two that recovered had 4058 and 4087 erasures and the one that failed had 4130, exactly the n − k = 4096 rule. But the true recovery probability at ε = 0.50 is 0.504, and three trials returned two successes, which happens about half the time. Interpolating between two grid points turned that into a third decimal place.
- **λ=3 came out at 0.275.** Its data jumped from 8/8 at 25% straight to 0/8 at 30%, with no trial in between, so the estimate was simply the midpoint of a five-point gap.

Each threshold is a **maximum-likelihood logistic fit** to every trial in that scheme's sweep, so a lucky rate is weighed against the rest rather than defining the answer, with a **95% bootstrap confidence interval** (`thresholds.py`). This replaces interpolation between the two rates bracketing 50%, which rests on just two points and gives no interval.

1D-RS is a check on the whole procedure, because its answer is known exactly: an MDS code recovers if and only if at most n − k symbols are erased, and on BEC(ε) that count is Binomial(n, ε). The sweep measures 0.499 (CI 0.497–0.501) against an exact 0.50006, and every one of its 132 trials obeys the n − k rule.

Code parameters and storage cost

All of them carry the same 124 KiB blob payload; they differ in how much they store to protect it. `R = k/n` is the code rate, and `1−R` is the largest erasure fraction *any* code of that rate could survive. The two rows the supervisor asked for — **No and Ko, the component (local) code length and dimension** — are included: for 1D-RS the global code is the only code, for 2D-RS the component is one axis (the row axis is shown; the column axis is [88, 64]), and for BC it is one local block.

Note the gap between the last two rows for 2D-RS. Its minimum distance is the *product* of its component distances, 33 × 25 = 825, so its guarantee is 0.098 — nowhere near the 0.515 its rate would permit. A product code is never MDS; that is the structural price of two-axis locality. Comparing 2D-RS against $1-R$, as the previous revision did, measured it against a bound its construction cannot reach.

	1D-RS	2D-RS	BC ω2048 λ2	BC ω1024 λ2	BC ω512 λ2	BC ω1024 λ3	BC ω1024 λ4
Encode	0.23s	1.36s	15.78s	6.88s	3.35s	7.72s	4.64s
Decode @ 2%	2.65s	1.27s	0.52s	0.29s	0.17s	0.37s	0.19s
Decode @ 5%	5.97s	1.60s	1.22s	0.65s	0.37s	0.84s	0.46s
Decode @ 10%	10.79s	1.72s	2.48s	1.30s	0.71s	1.68s	0.91s
Decode @ 20%	20.50s	2.46s	5.43s	2.77s	1.48s	3.57s	1.86s
Decode @ 30%	28.50s	2.34s	9.01s	4.59s	2.43s	—	—
Decode @ 35%	32.63s	3.00s	—	—	2.77s	—	—
Decode @ 48%	41.62s	—	—	—	—	—	—

Results

Recovered trials at each erasure rate. Trial counts differ by scheme and by rate — more where a scheme is near its own threshold and the outcome genuinely varies, fewer where it does not — so each cell shows

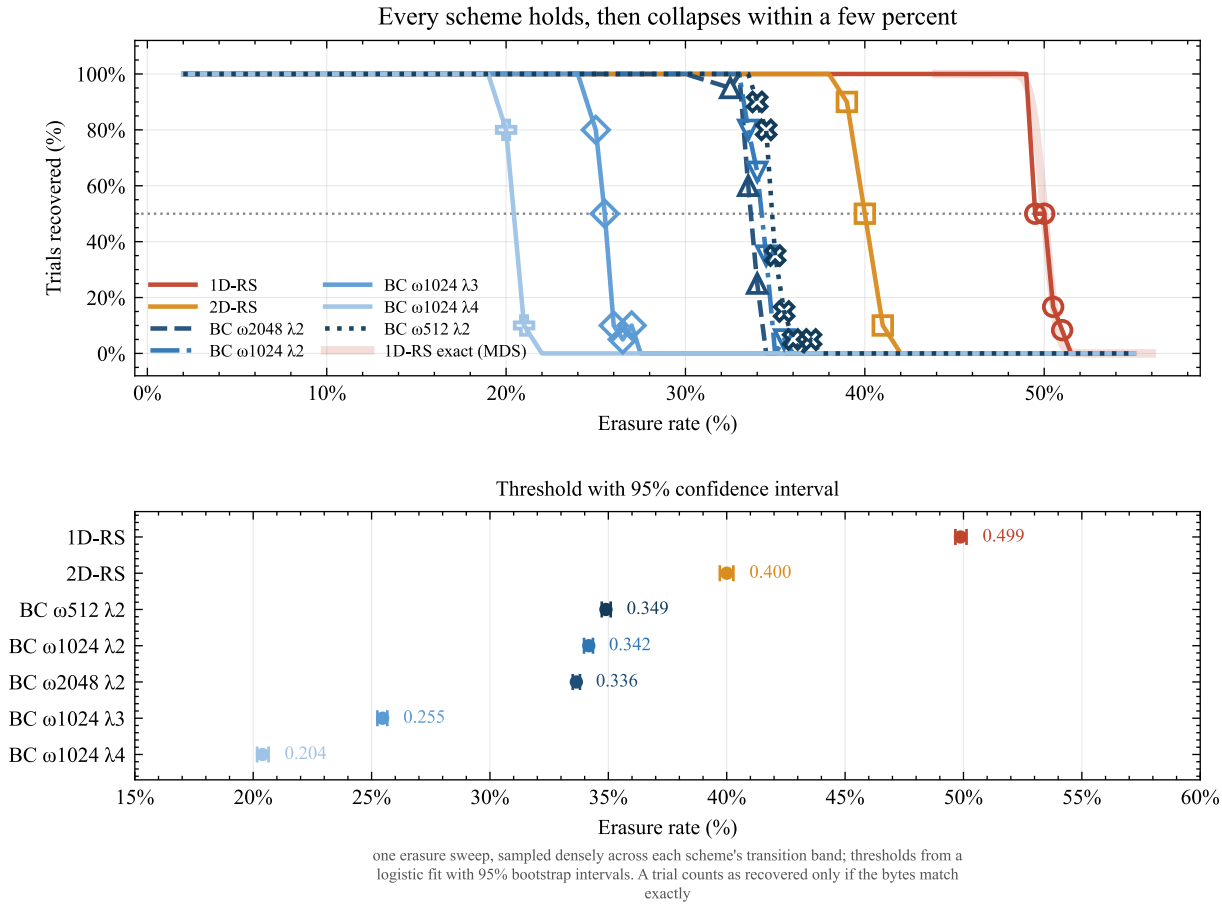
recovered/attempted rather than a bare percentage.

erasure rate	1D-RS	2D-RS	BC ω2048 λ2	BC ω1024 λ2	BC ω512 λ2	BC ω1024 λ3	BC ω1024 λ4
2%	1/1	2/2	2/2	2/2	2/2	2/2	2/2
5%	1/1	2/2	2/2	2/2	2/2	2/2	2/2
8%	1/1	2/2	2/2	2/2	2/2	2/2	2/2
10%	1/1	2/2	2/2	2/2	2/2	2/2	2/2
12%	1/1	2/2	2/2	2/2	2/2	2/2	2/2
15%	1/1	2/2	2/2	2/2	2/2	2/2	2/2
16%	1/1	2/2	2/2	2/2	2/2	2/2	20/20
17%	1/1	2/2	2/2	2/2	2/2	2/2	20/20
18%	1/1	2/2	2/2	2/2	2/2	2/2	20/20
19%	1/1	2/2	2/2	2/2	2/2	2/2	20/20
20%	1/1	2/2	2/2	2/2	2/2	2/2	16/20
21%	1/1	2/2	2/2	2/2	2/2	2/2	2/20
22%	1/1	2/2	2/2	2/2	2/2	2/2	0/20
23%	1/1	2/2	2/2	2/2	2/2	2/2	0/20
24%	1/1	2/2	2/2	2/2	2/2	2/2	0/20
25%	1/1	2/2	2/2	2/2	2/2	16/20	0/20
25.5%	1/1	2/2	2/2	2/2	2/2	10/20	0/2
26%	1/1	2/2	2/2	2/2	2/2	2/20	0/2
26.5%	1/1	2/2	2/2	2/2	2/2	1/20	0/2
27%	1/1	2/2	2/2	2/2	2/2	2/20	0/2
27.5%	1/1	2/2	2/2	2/2	2/2	0/20	0/2
28%	1/1	2/2	2/2	2/2	2/2	0/20	0/2
28.5%	1/1	2/2	2/2	2/2	2/2	0/20	0/2
29%	1/1	2/2	2/2	2/2	2/2	0/20	0/2
29.5%	1/1	2/2	2/2	2/2	2/2	0/20	0/2
30%	1/1	2/2	2/2	2/2	2/2	0/20	0/2
32.5%	1/1	2/2	19/20	20/20	2/2	0/2	0/2
33%	1/1	2/2	20/20	20/20	20/20	0/2	0/2
33.5%	1/1	2/2	12/20	16/20	20/20	0/2	0/2
34%	1/1	2/2	5/20	13/20	18/20	0/2	0/2
34.5%	1/1	2/2	0/20	7/20	16/20	0/2	0/2
35%	1/1	2/2	0/20	0/20	7/20	0/2	0/2
35.5%	1/1	2/2	0/20	1/20	3/20	0/2	0/2

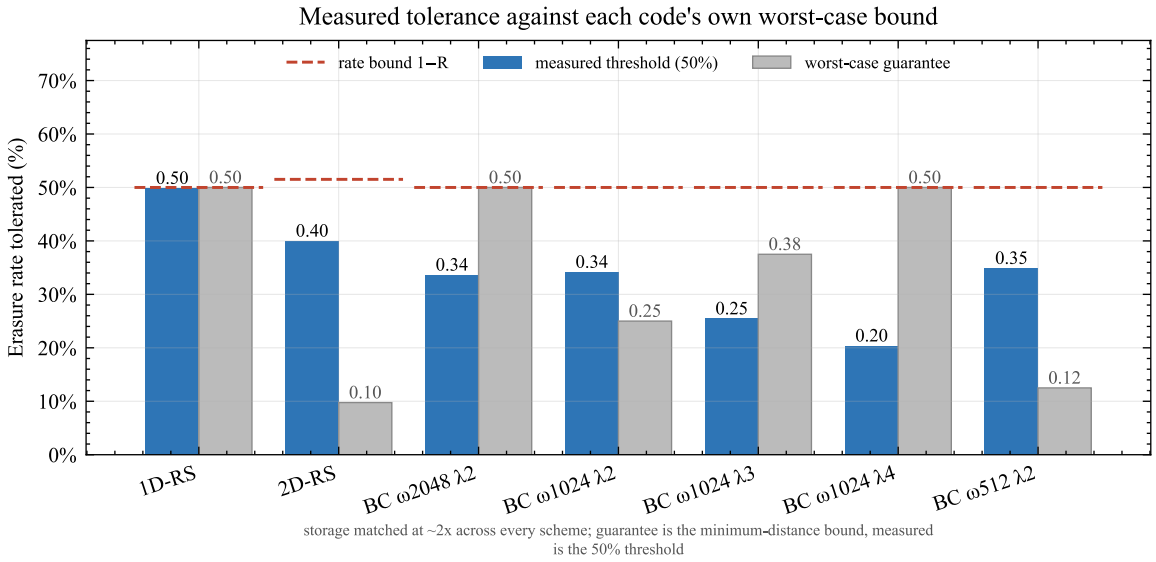
erasure rate	1D-RS	2D-RS	BC ω2048 λ2	BC ω1024 λ2	BC ω512 λ2	BC ω1024 λ3	BC ω1024 λ4
36%	1/1	2/2	0/2	0/20	1/20	0/2	0/2
36.5%	1/1	2/2	0/2	0/20	0/20	0/2	0/2
37%	1/1	20/20	0/2	0/2	1/20	0/2	0/2
37.5%	1/1	20/20	0/2	0/2	0/20	0/2	0/2
38%	1/1	20/20	0/2	0/2	0/2	0/2	0/2
39%	1/1	18/20	0/2	0/2	0/2	0/2	0/2
40%	1/1	10/20	0/2	0/2	0/2	0/2	0/2
41%	1/1	2/20	0/2	0/2	0/2	0/2	0/2
42%	1/1	0/20	0/2	0/2	0/2	0/2	0/2
43%	1/1	0/20	0/2	0/2	0/2	0/2	0/2
44%	1/1	0/20	0/2	0/2	0/2	0/2	0/2
45%	1/1	0/2	0/2	0/2	0/2	0/2	0/2
48%	1/1	0/2	0/2	0/2	0/2	0/2	0/2
48.5%	12/12	0/2	0/2	0/2	0/2	0/2	0/2
49%	12/12	0/2	0/2	0/2	0/2	0/2	0/2
49.5%	6/12	0/2	0/2	0/2	0/2	0/2	0/2
50%	6/12	0/2	0/2	0/2	0/2	0/2	0/2
50.5%	2/12	0/2	0/2	0/2	0/2	0/2	0/2
51%	1/12	0/2	0/2	0/2	0/2	0/2	0/2
51.5%	0/12	0/2	0/2	0/2	0/2	0/2	0/2
52%	0/1	0/2	0/2	0/2	0/2	0/2	0/2
55%	0/1	0/2	0/2	0/2	0/2	0/2	0/2

Every block-circulant row here is decoded by the Vandermonde solver, not the NTT. The section on the NTT decoder below measured both paths and found the Vandermonde solver faster by a wide margin, and the decoder's phase 1 now uses it alone, so there is no switch left to set. The erasure tolerances are identical either way; only the decode times depend on it.

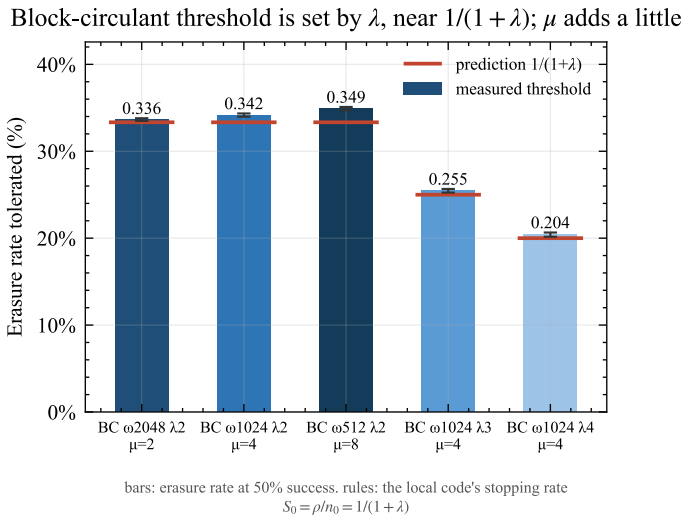
Across all 1,800 trials, no scheme returned corrupted data while reporting success: silent corruption was zero everywhere (0/132 for 1D-RS, 0/272 for 2D-RS, and 0/236, 0/272, 0/290, 0/308, 0/290 across the five BC rows in the table's order). Before the decoder fix described below, BC's figure was 31 of 40.



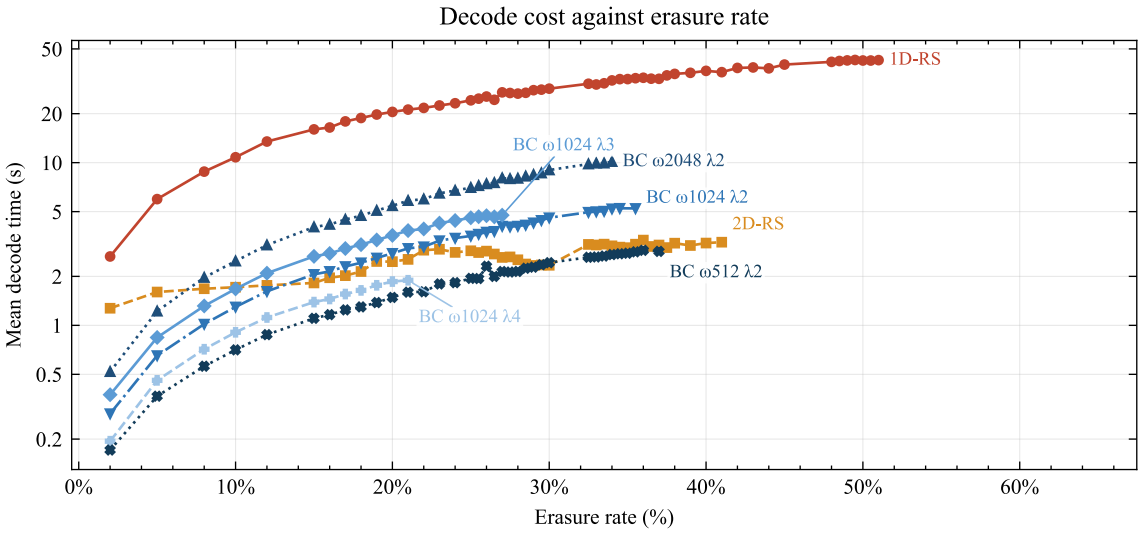
Fault tolerance of all seven schemes. **(a)** Fraction of trials recovered against erasure probability, from the 1,800-trial sweep, which samples each scheme's transition band densely. Every scheme holds at 100% and then collapses within a few percent, which is what makes a single threshold a fair summary. The pale band is the exact recovery probability of 1D-RS, computed from the MDS property rather than fitted; the measured points sit on it. **(b)** Each threshold — the erasure probability at which a scheme recovers half the time — with its 95% bootstrap confidence interval. The three λ=2 rows land within 0.013 of one another (0.336, 0.342, 0.349), far closer than to λ=3 or λ=4, so λ sets the reach; but their intervals do not overlap and they rise with μ, so μ has a small, real effect of its own.



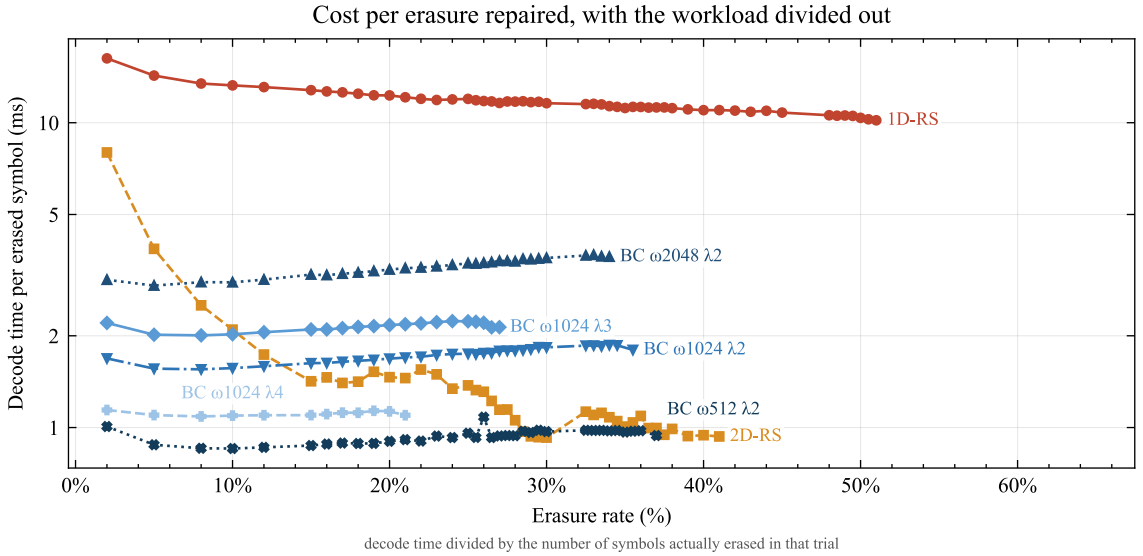
What each scheme actually tolerated (blue), beside its own minimum-distance guarantee (grey) and the rate bound $1-R$ that no code of that rate can beat (dashed red). Storage is matched to within 3%, so the bars read against each other directly. 1D-RS lands on its bound — 0.499 measured, against an exact 0.50006 for an MDS code on an erasure channel — which is what being MDS means. Three schemes sit *above* their own guarantee: 2D-RS at four times it, and both λ=2 block-circulant rows with μ>2. That is not a contradiction — a guarantee holds against every pattern including the adversarial worst case, while these trials draw patterns at random. Each blue bar is the erasure probability at which the scheme recovers half the time, from a maximum-likelihood logistic fit to every trial in the sweep (`thresholds.py`).



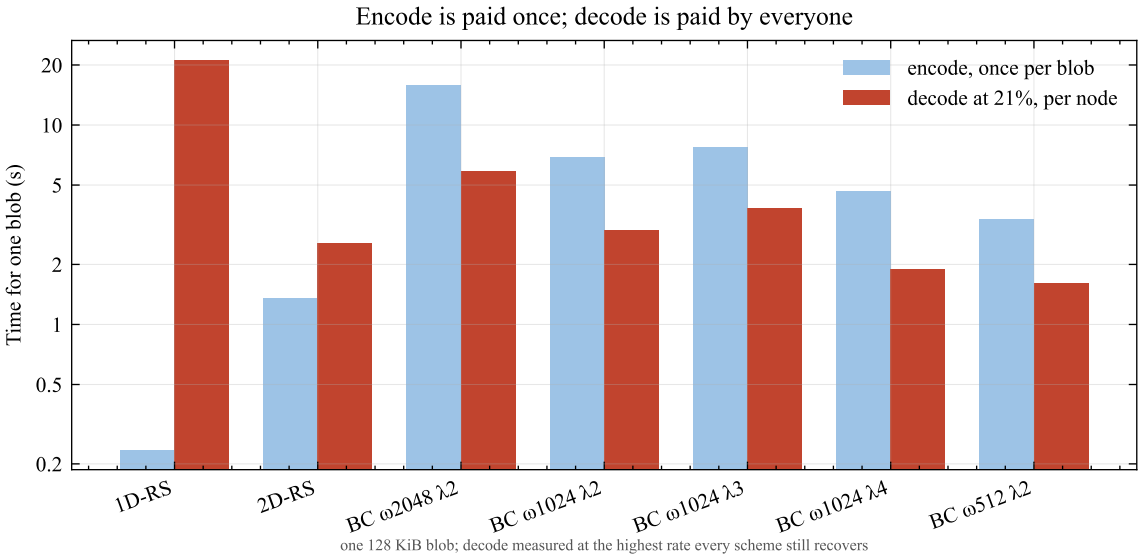
The block-circulant parameter sweep against the source paper's prediction. Bars are measured thresholds with 95% confidence intervals, ordered by λ and then by μ; the red rules are the local stopping rate So = ρ/No = 1/(1+λ). λ dominates: holding μ=4 fixed, raising λ from 2 to 3 to 4 takes the threshold from 0.342 to 0.255 to 0.204. Every measurement sits just above its prediction, which is the expected direction — So is what one block can repair alone, and the cascade across blocks can only add to it. That also explains the smaller effect at fixed λ=2: the threshold rises from 0.336 to 0.342 to 0.349 as μ goes from 2 to 4 to 8, because more blocks give the cascade more to work with.



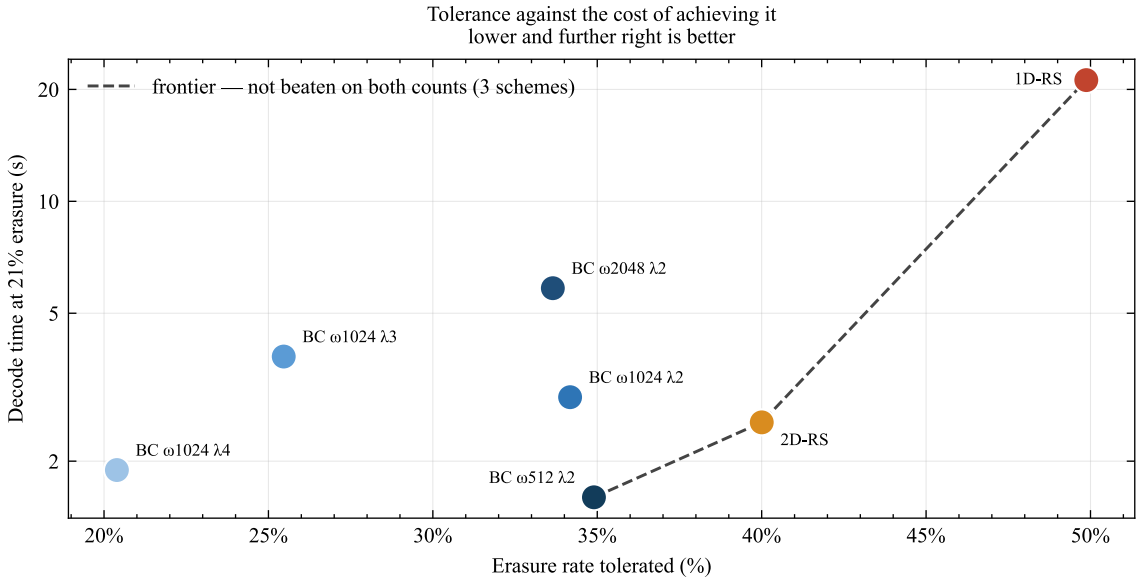
Mean decode time of the trials that actually recovered, against erasure rate, log scale. Failed trials are excluded: above its threshold a scheme rejects the pattern on a budget check in under a millisecond having attempted nothing, which is not a decode time. 1D-RS is the most expensive scheme at every rate, by a factor of three to five over the dearest block-circulant row, because it divides one 8192-length polynomial while the others divide short ones. Every block-circulant row is cheaper than 1D-RS at every rate measured. Against 2D-RS the picture is now mixed rather than one-sided: below about 7% erasure every BC row is cheaper, but 2D-RS decode barely grows with erasure rate while BC decode grows roughly linearly, so the rows cross it one by one — ω2048 λ2 first, then ω1024 λ3 and ω1024 λ2 by about 13% — and only ω512 λ2 and ω1024 λ4 stay at or below it across their whole range. These are Vandermonde-path timings, with the NTT path switched off — see the NTT section for what turning it on costs, and for why defining the local code over roots of unity raises the block-circulant numbers here by roughly a factor of two whether or not the transform is ever used.



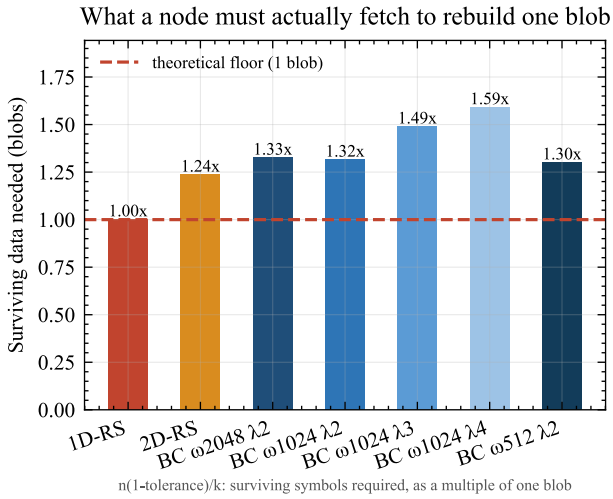
The same decode times with the workload divided out — cost per erasure actually repaired, rather than cost per decode. Raw decode time rises with erasure rate for every scheme, which mostly says there is more to fix; dividing by the erasure count removes that and leaves the algorithmic difference. 1D-RS sits near 11–13 ms per erasure over most of the range — 16 ms at 2%, where a fixed overhead is spread over few erasures, easing to 10.5 ms near its threshold: its cost is essentially linear in the number of erasures, one long division's worth of work each. The block-circulant rows are three to sixteen times cheaper per erasure (ω2048 λ2 least, ω512 λ2 most) and rise gently, which is the short local divisions doing their job. 2D-RS is the interesting shape — it starts around 8 ms and falls steeply toward 1 ms, because its decode carries a large fixed cost that amortises as more cells need repairing. That is why it looks expensive at 2% erasure and cheap at 30%.



Encode and decode are not symmetric in a data-availability system and should not be collapsed into one cost. A blob is encoded *once*, by the proposer; it is decoded by every node that has to reconstruct it, so an expensive encode amortises across the network while an expensive decode does not. The two Reed-Solomon schemes and the block-circulant family sit on opposite sides of this: 1D-RS encodes in under a quarter of a second and then costs 21 s to decode, while BC ω2048 λ2 spends 16 s encoding to decode in 6 s (decode measured at 21%, the highest rate every scheme still recovers). For PeerDAS the second shape is the more useful one, which is an argument for the block-circulant family that the tolerance numbers alone do not make.



The trade-off the whole comparison is about, in one picture: how much erasure a scheme survives against what it costs to decode at a rate every scheme reached. Down and to the right is better. The dashed line is the Pareto frontier — the schemes not beaten on both counts at once — and only three schemes are on it. BC ω512 λ2 is the cheapest thing that still tolerates a third of the codeword; 2D-RS buys about five more points of tolerance for about 60% more decode time; 1D-RS buys the theoretical maximum and pays an order of magnitude for it. Everything else is dominated: some other scheme tolerates more erasure *and* decodes faster. Note especially that BC ω1024 λ4 is dominated by BC ω512 λ2 on both axes, so the tolerance it gives up buys nothing back.



How much surviving data each scheme needs to rebuild one blob, as a multiple of the blob itself. This is the question that matters for data-availability sampling, where a node has to actually fetch those symbols. 1D-RS needs essentially the theoretical floor — no code rebuilds 124 KiB from less than 124 KiB. Every other scheme needs more, and a scheme tolerating a lower erasure rate at the same storage cost necessarily needs more.

What the results show

1D-RS behaves exactly as Reed-Solomon theory says it must. Its threshold is **0.499** (95% CI 0.497–0.501), against an exact 0.50006 for an MDS code with $n = 2k$ on an erasure channel. It recovers every trial up to 49% erasure and none from 51.5%. The invariant underneath is exact rather than approximate: across all 132 trials, recovery succeeded if and only if the drawn erasure count was at most $n-k = 4096$, with no exceptions. The partial results at the threshold are not the decoder being unreliable — they are the random mask landing on either side of a hard line. That also accounts for the point estimate sitting a hair under the exact value: at $\varepsilon = 0.495$, where the exact recovery probability is 0.82, six of the twelve masks happened to carry more than 4096 erasures — a draw with about a 1% chance — and each of those six failed, exactly as the rule says it must. Being MDS, 1D-RS extracts all of its theoretical budget.

2D-RS comes second, and beats its own guarantee by a wide margin. Its threshold is **0.400** (95% CI 0.397–0.403). Two reference points matter and they are very far apart. Its rate would permit 0.515; its *minimum distance* only guarantees 0.098, because a product code's distance is the product of its component distances, $33 \times 25 = 825$. The measured 0.400 is four times that guarantee.

That gap is the cross-axis cascade doing real work. A guarantee has to hold against the worst pattern in existence — for a product code, an adversary who concentrates erasures to kill whole rows and columns at once. Random patterns are nothing like that, so lines that are individually over budget keep getting rescued once their partner axis repairs enough neighbours. The honest summary is that 2D-RS's *typical* behaviour is far better than its *guaranteed* behaviour, and which of the two matters depends entirely on whether the erasures are adversarial. Against the $1-R$ bound it reaches 78%; against its own distance it is at 410%.

The Block-Circulant rows separate the code from the decoder, and λ is what moves the decoder. This is the clearest result in the sweep:

row	μ	*d*	threshold	95% CI	predicted s_0
ω2048 λ2	2	4097 (MDS)	**0.336**	0.335–0.338	0.333
ω1024 λ2	4	2049	**0.342**	0.340–0.343	0.333
ω512 λ2	8	1025	**0.349**	0.347–0.351	0.333
ω1024 λ3	4	3073	**0.255**	0.252–0.257	0.250
ω1024 λ4	4	4097 (MDS)	**0.204**	0.202–0.206	0.200

Two things fall out of that table. First, **the prediction holds every time, from just below.** $s_0 = \rho/N_0 = 1/(1+\lambda)$, the source paper's local stopping rate, sits under the measured threshold for all five rows, by between 0.003 and 0.016. That is the right side to be on: s_0 is what one block can repair on its own, and the cascade across blocks can only add to it. The limit is not mysterious; it is arithmetic about how many erasures fit inside one local support.

Second, **λ is the parameter that moves it, and μ moves it slightly.** Holding $\mu = 4$ fixed, raising λ from 2 to 3 costs 9 points and to 4 costs 14. Holding $\lambda = 2$ fixed, quadrupling μ from 2 to 8 *gains* just over a point — 0.336, 0.342, 0.349 — a small effect but a real one, since the three intervals do not overlap. It has the same cause as the gap above s_0: more blocks give the cross-block cascade more to work with. The answer to "which knob should we turn" is still λ, and downward; μ is a second-order refinement.

Third, and most useful for the construction's prospects, **two of these rows decode past their own worst-case guarantee.** ω1024 λ2 is guaranteed only 0.250 and reaches 0.342; ω512 λ2 is guaranteed 0.125 and reaches 0.349, nearly three times its bound. These are *not* MDS codes and cannot claim to be; against a worst-case adversary they would fail where the table says they fail. But under random erasure — which is what a sampling network actually produces absent a targeted attack — they perform like codes with far better distance. That is the sense in which parameters that are "not fully decodable" still demonstrate the construction's potential.

Set against that, every BC row is cheaper than 1D-RS at every rate measured. Within the block-circulant family the ordering is now set as much by the field as by λ: ω1024 λ4 decodes faster than the λ=2 and λ=3 rows of the same ω, not because λ=4 is a better choice but because its local length 5120 does not divide $p-1$, so it never moved onto 255-bit evaluation points and never paid for them. The NTT section below measures that cost. The local system each solves is $\rho \times \rho$, which generic elimination handles in ρ^3 — which is why the pairwise phase stays unaffordable while the single-block phase does not, the latter's system being Vandermonde and so dropping to ρ^2.

Making these parameterisations runnable

The larger-ω BC rows could not be measured at all with the decoder as it stood. The reason is worth recording, because it is the same wall anyone re-running this will hit.

The original `gf_solve` is a Gauss-Jordan elimination written as a Python loop over rows. Its cost grows as the cube of the matrix dimension, and this field's arithmetic runs in `galois`' object mode at roughly 1.3 microseconds per element operation. At ω=ρ=2 the local systems are 2×2 and none of that matters. At ω=ρ=1024 they are 1024×1024, which is about a billion element operations — roughly 25 minutes for a single local solve, with μ of them per decoding round.

Three changes brought it into range, none of which alters what is computed:

The elimination step is now one broadcast update instead of a loop over rows. Same arithmetic, same element count, but the per-pivot dispatches into object-mode arithmetic drop from m to one. Measured: 1.34 → 0.55 microseconds per element operation, a 2.4× speedup.

The single-block solve now exploits the structure it always had. The local parity-check matrix is a GRS matrix, `H[r, c] = (c+1)^r`. Restricted to the erased columns and to its first e rows it is a square Vandermonde in distinct evaluation points — so it is always invertible, and it can be solved in $O(e^2)$ by a node-polynomial argument rather than $O(e^3)$ by elimination. Measured at $e = 1024$: 0.87 seconds against an extrapolated 1441, about 1650×. This is what makes $\omega=1024$ tractable.

The global parity-check matrix is no longer built. It is $(\mu\rho)\times n = 4096\times 8192 = 33.5$ million field elements, and every row the decoder needs can be regenerated from (ρ, n_0) on demand.

A useful side effect: because the single-block system is provably square Vandermonde in distinct nodes, it cannot be underdetermined, so the class of bug described below cannot arise on that path at all.

The pairwise phase does not get a turn

`DecCBC[2]`'s second phase stacks two adjacent blocks' parity checks and solves them jointly. That system is *not* Vandermonde — the two row groups assign different evaluation points to the same shared columns — so it cannot use the fast solver and needs full elimination, at cubic cost.

Phase 2 only runs when phase 1 has stalled, which happens only once a local support carries more than ρ erasures. The sweep applies a stated compute budget of 256 unknowns per pairwise system, identically to all five rows, and above that budget a pair is skipped, counted, and its positions left honestly flagged. Every row here has $\rho \geq 512$, so a pair facing the *full* erasure load is always over budget.

What that could miss is the residual: once phase 1 has resolved some blocks, a pair can in principle be left with few enough unknowns to fall under the budget. Whether that ever happened was checked directly rather than assumed. Every one of the 1,396 block-circulant trials was replayed with the decoder's phase counters switched on, the masks regenerated from the sweep's seed; every erasure count and every outcome reproduced exactly. **Phase 2 never ran.** In 674 of those trials phase 1 stalled with a pair over the budget, which was skipped; in the rest phase 1 either finished the job or stalled with nothing a pair could take on. So **all five rows are measured as single-phase decoders** — exactly, not approximately — and every block-circulant threshold above is what the single-block phase alone can reach. That is a limit of affordable computation rather than of the code.

The replay also settles the timings. Decode times are wall-clock, so a machine that slept or throttled mid-decode would inflate them. None of the 1,800 trials shows that: no successful decode took more than 1.15 times the median of the same scheme's decodes within three points of erasure rate, and the longest decode of all was 44 s, an ordinary 1D-RS decode near its threshold.

Since the decoder was revised, phase 2 is the paper's Algorithm 1 auxiliary construction only; the earlier Gaussian-elimination fallback on the stacked parity checks was removed. Algorithm 1 also needs the paper's evaluation-point layout, which the benchmark's encoder does not use, so even a pair under the budget would have had little to gain. The cost of the earlier approach, measured directly on the 512-equation systems that $\omega=\rho=256$ produces, shows why phase 2 was never affordable at full load:

unknowns in the pair	erasure rate it corresponds to	per pair	per decoding round (16 pairs)
276	6%	21.7s	5.8 min
369	8%	39.5s	10.5 min
461	10%	61.7s	16.4 min

That is for ω=ρ=256 — smaller than any local block in the current sweep; the measurement dates from an earlier parameter set with sixteen blocks, hence sixteen pairs per round — and one decoding round is not one decode: the loop repeats while it makes progress. At ω=ρ=1024 a full pair is 2048 equations in 2048 unknowns; at the 0.57 microseconds per field operation these timings imply, that is about 80 minutes for a single pair. Lifting the cap enough to change any result would take days per sweep.

The honest reading of the BC numbers is therefore: this is what the single-block phase of `DecCBC[2]` achieves. A working pairwise phase would roughly double the budget from ρ to 2ρ, spread over the union of two adjacent supports — 2ρ + min(λ+1, μ)·ω symbols. As a plain budget count, which ignores how erasures actually fall across pairs and so is an upper estimate, that puts the reach at 0.50 for ω2048 λ2, 0.40 for ω1024 λ2 and ω512 λ2, and 0.333 for λ3 and λ4, against the single-block reach of 1/(1+λ) — 0.333, 0.250 and 0.200 — that the measured thresholds sit just above. Only ω2048 λ2, whose two blocks share every information block, would reach the 0.50 these codes permit.

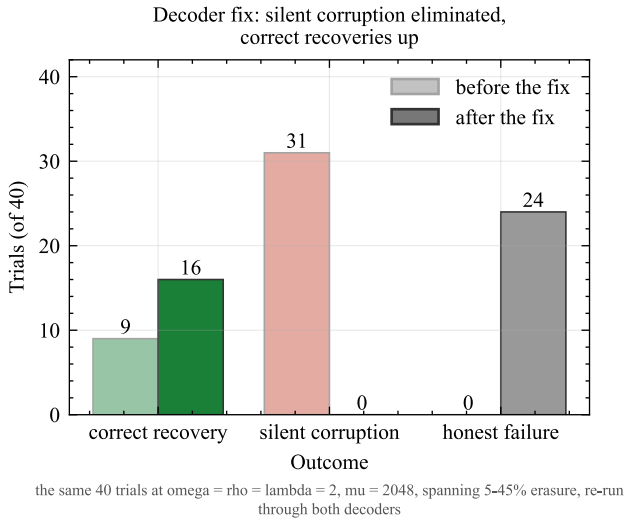
The Block-Circulant decoder fix

The BC results are from a corrected decoder. The version originally benchmarked had two faults that together made its output untrustworthy rather than merely weak. Both were found at the earlier ω=ρ=λ=2 parameterisation, and the before/after figure below is measured there.

`gf_solve` fabricated values. The Gaussian elimination routine returned zero for any unknown whose column never received a pivot. To the caller that is indistinguishable from having genuinely solved for zero, so an underdetermined system silently produced a confident-looking wrong answer. It now raises `GFSolveError` unless the system has exactly one solution — every column pivoted, and no row reduced to `0 = nonzero`.

`dec_cbc_2`'s pairwise phase accepted systems it had not solved. Its guard is `0 < erasures ≤ 2ρ`, which is necessary but not sufficient. The equations it stacks are block *i*'s own ρ parity checks on top of block *i+1*'s — not the auxiliary `[2(ω+ρ), 2ω, 2ρ+1]` GRS code over the union of the two supports that the source paper's Algorithm 1 constructs. The stacked form is not MDS, so a split such as "three erasures inside block *i*'s support, one inside *i+1*'s" leaves two equations facing three unknowns. Combined with the `gf_solve` fault, the decoder wrote zeros into those positions *and cleared their erasure flags*, reporting success on corrupted data. It now skips any pair it cannot uniquely solve, leaving those positions honestly flagged.

The practical effect is stark. Measured before the fix, across 40 trials spanning 5–45% erasure, `dec_cbc_2` claimed success **every single time** and was wrong on 31 of them; it never once reported a failure. Its `remaining` flag — the signal a caller would naturally trust — carried no information about whether the data was correct. Had this benchmark trusted that flag instead of checking bytes, it would have reported BC at 100% success at every rate.



The decoder fix, measured on the parameterisation it was found at (ω=ρ=λ=2, μ=2048) — *not* the rows swept above. The same 40 trials (5%, 15%, 25% and 45% erasure, ten each) run through the decoder before and after. Correct recoveries rose from 9 to 16, because the old decoder poisoned positions with fabricated zeros that blocked later rounds from resolving them. Silent corruptions — cases where the decoder reported success on wrong data — fell from 31 to zero. The remaining failures are honest: the decoder reports the positions it could not resolve.

A third change is a performance fix rather than a correctness one: the pairwise phase used to `break` and restart the single-block phase after every pair it repaired, paying a full scan of all local blocks per repair. It now continues through the remaining pairs in the same pass.

What none of this does is restore the correction radius the paper claims. Implementing the auxiliary GRS construction would let the pairwise phase handle *any* $\leq 2\rho$ erasures across two adjacent blocks, rather than only those solvable from the stacked parity checks — and, being a GRS code, it would also be solvable in $O(e^2)$ rather than $O(e^3)$, which would remove the compute wall described above. That is the single highest-value piece of remaining work on this file.

An implementation penalty the block-circulant rows carry

One caveat belongs with the decode times above, because it affects the comparison between schemes rather than just their absolute values.

Making the block-circulant local code FFT-capable means defining it over roots of unity, since that is what an NTT requires. But the evaluation points *define the code* — the encoder uses the same ones — so the Vandermonde solver ends up working over 255-bit field elements where the earlier version used the small integers 1..no. In pure-Python modular arithmetic operand width is not free, and **this cost is paid whether or not the transform ever runs**.

Measured across three versions of the decoder on the same machine, same erasure patterns, min of three (2.33× median):

variant (20% erasure)	small-integer points	roots of unity	penalty
BC ω2048 λ2	2.27s	5.33s	2.34×
BC ω1024 λ2	1.20s	2.74s	2.28×
BC ω1024 λ3	1.50s	3.54s	2.37×
BC ω512 λ2	0.66s	1.43s	2.15×

The penalty falls **only on the block-circulant rows**. 1D-RS and 2D-RS are untouched by it, and so is BC ω1024 λ4, whose local length $5120 = 2^{10} \cdot 5$ does not divide $p-1$ and which therefore never moved onto roots of unity.

Two consequences for reading the decode figures. The gap between 1D-RS and the block-circulant family is **narrower than it would otherwise be** — about three to five times rather than an order of magnitude. And the crossover with 2D-RS, which the decode-cost figure places near 7% erasure, would sit substantially higher without the penalty. Neither changes any tolerance result: success rates are identical across all three versions, verified cell by cell.

This is a property of this implementation, not of block-circulant codes. `decoder_version_benchmark.py` reproduces the measurement.

The NTT decoder path

`BCcodeEthereum.py` carries a second decoder for the single-block phase, built around a number-theoretic transform rather than the Vandermonde solver described above. The motivation is sound: the source paper claims $O(n \log^2 n)$ for this step, the local code is a GRS code, and a GRS erasure decode is a transform, a pointwise multiply and a polynomial division. On paper it should beat an $O(e^2)$ solve comfortably.

It does not, and the reasons are worth recording, because two of the three are not about the transform at all.

It was not running

The first finding is that the NTT path had never once executed to completion. Every call ended in the same guard —

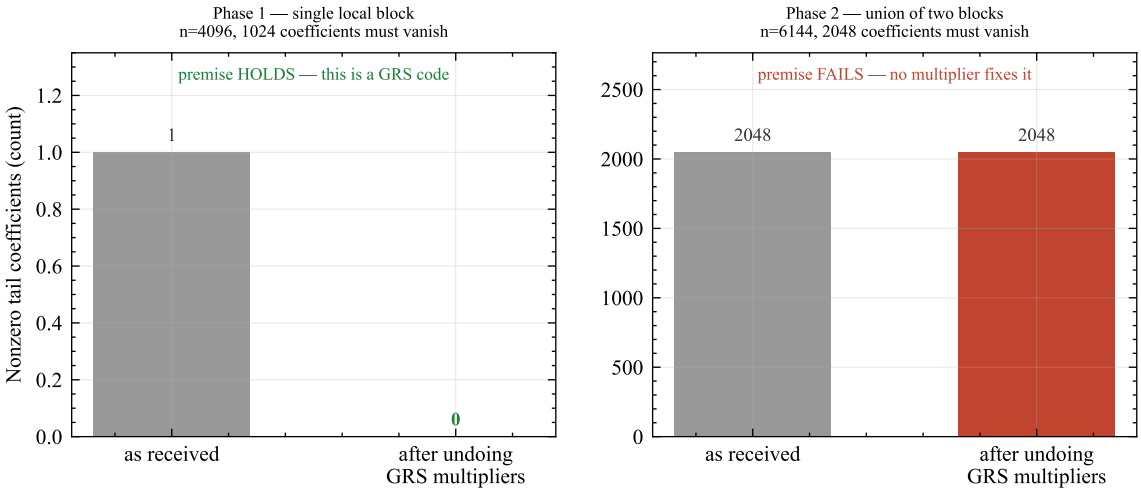
```
FFT failed (nonzero tail coefficients: input not a valid codeword)
```

— and fell through to the Vandermonde solver. The benchmark numbers were therefore *Vandermonde numbers plus the cost of a failed NTT attempt*, and the NTT rows looked slower for a reason that had nothing to do with the transform.

The guard itself is correct and worth keeping. `fft_grs_erasure_decode` inverse-transforms the locator-multiplied received word and requires the coefficients above $k + e$ to vanish; if they do not, the input is not a codeword of the code it assumed, and returning a confident wrong answer would be far worse than bailing out. The premise was what was wrong.

The local code is not a set of bare polynomial evaluations. `build_local_grs` defines it by the parity check $H[r, c] = x_c^r$, which makes it a *generalised* Reed-Solomon code: its codewords are $c_j = v_j \cdot f(x_j)$ with column multipliers $v_j = 1 / \prod_{i \neq j} (x_j - x_i)$. For no-th roots of unity that product is no / x_j, so v_j is proportional to x_j. The decoder was handed codewords and treated them as evaluations, which they are not.

Dividing the received word through by x_c before transforming, and multiplying back afterwards, fixes it exactly — not approximately.



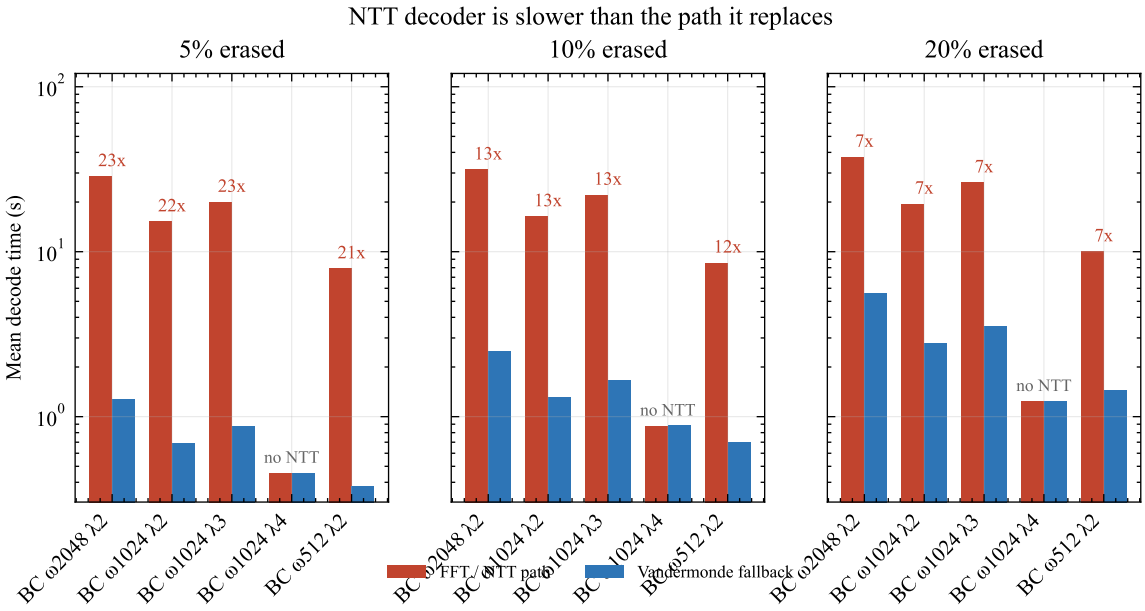
The premise the NTT decoder rests on, measured as the number of polynomial coefficients that are required to vanish and do not. **Phase 1** starts with a single stray coefficient and lands on exactly zero once the column multipliers are divided out — the local code really is GRS, and the fix is the right one. **Phase 2** is a different situation entirely: every one of its 2048 tail coefficients is nonzero, and stays nonzero after the same correction. No choice of multipliers repairs it, because the problem is not multipliers. Generated by `fft_benchmark.py`; the same two checks run as assertions in `test_bc_fft.py`, where the phase-2 case is asserted to *fail* so that the test turns red if the auxiliary construction is ever implemented.

Now that it runs, it is slower

With the multipliers applied the NTT path decodes correctly on every FFT-eligible parameterisation. It is also between about seven and twenty-three times slower than the solver it was meant to replace.

variant	no	NTT usable	5%	10%	20%
BC ω2048 λ2	6144	yes	28.7s vs **1.3s** — 23×	31.5s vs **2.5s** — 13×	37.6s vs **5.6s** — 7×
BC ω1024 λ2	3072	yes	15.3s vs **0.7s** — 22×	16.4s vs **1.3s** — 13×	19.6s vs **2.8s** — 7×
BC ω1024 λ3	4096	yes	20.1s vs **0.9s** — 23×	22.2s vs **1.7s** — 13×	26.3s vs **3.6s** — 7×
BC ω512 λ2	1536	yes	7.9s vs **0.4s** — 21×	8.6s vs **0.7s** — 12×	10.1s vs **1.5s** — 7×
BC ω1024 λ4	5120	no	**0.5s** (fallback only)	**0.9s** (fallback only)	**1.2s** (fallback only)

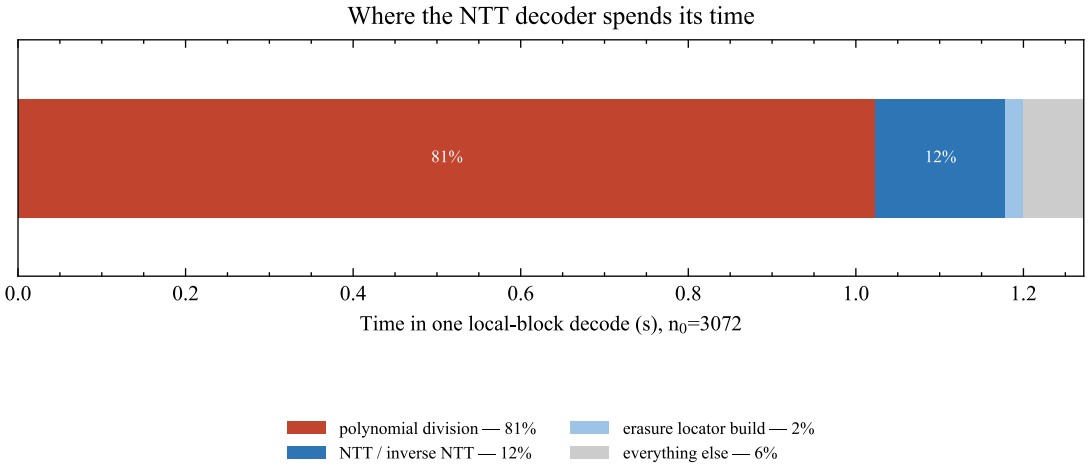
Each cell is the NTT path against the Vandermonde fallback, mean of 3 trials, both decoding the same corrupted codeword in the same process. **Bold is the faster path.** Phase 2 is held off throughout (`use_fft_phase2=False`), so this isolates the single-block phase. This was measured on the decoder version that still exposed the NTT switch; the current decoder's phase 1 is Vandermonde-only, so `fft_benchmark.py` carries this measurement forward rather than repeating it.



The same corrupted codeword decoded twice in the same process, once down each path, so machine speed and field-construction cost cancel. The NTT path loses everywhere it is available. It does scale better — across this rate range its cost grows by about a third while the Vandermonde solver's roughly quadruples — but extrapolating that trend to BC ω2048 λ2's actual tolerance ceiling of 0.33 leaves it still roughly three to four times behind, so on this evidence the two never cross anywhere the decoder can actually operate. That last step is an extrapolation beyond the measured points and is offered as such. BC ω1024 λ4 is marked *no NTT*: its local length is $5120 = 2^{10} \cdot 5$, and the BLS12-381 scalar field has no element of order 5, so both bars are the same fallback.

The transform is not where the time goes

Profiling one local-block decode explains it. The NTT is a small minority of the work; the dominant cost is the polynomial division that follows it.



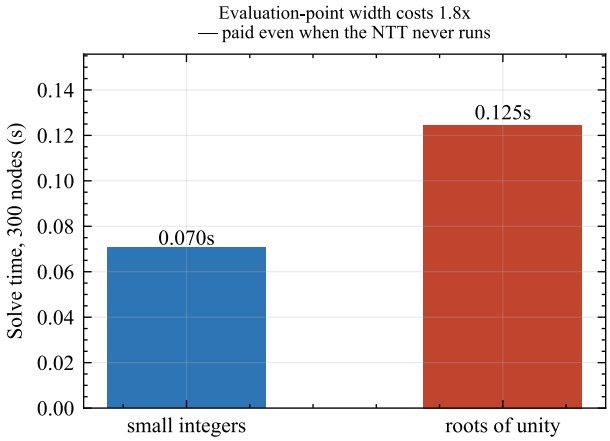
Where one local-block decode actually spends its time, measured by wrapping the three `galois` entry points the decoder calls rather than by bucketing a profiler dump — `cProfile` attributes the transform's work to a generic internal dispatcher, which makes the NTT look like it costs nothing at all. Roughly four fifths of the decode is a single `divmod` of two polynomials: `galois` implements that as a dense scalar loop, $O(n \cdot e)$ field operations in Python, which for these sizes is close to a million big-integer multiplications. The function is, in effect, an NTT wrapper around a schoolbook division.

This is a fixable problem, and the file's own docstring already names the fix: "*Replacing the divmod with Newton iteration gives the paper's claimed O(n log² n) bound.*" Until that is done, the asymptotic argument for the NTT path does not reach the code, because the step that dominates it is still quadratic.

Being FFT-capable is not free, even with the FFT switched off

The third cost is the least obvious and the most consequential for the numbers in this report.

An NTT needs its evaluation points to be roots of unity. But the evaluation points *define the code* — the encoder uses the same ones — so making the local code FFT-capable means defining it over 255-bit field elements instead of the small integers 1..*n*o the pre-NTT version used. The Vandermonde fallback then has to solve over those same 255-bit values, and in pure-Python modular arithmetic operand width is not free.



`vandermonde_solve` given identical input on 300 nodes, differing only in how wide those nodes are. Small integers 1..300 are 9 bits; roots of unity in this field are 253–255 bits. The solver does the same number of operations either way — the gap is the cost of the operands themselves.

An earlier run of the same comparison against the pre-NTT decoder (commit `8d93ae3`), on the same machine within the same hour, with the NTT path switched *off* so that both versions ran the same algorithm, also timed the λ4 row:

variant	pre-NTT	current	
BC ω2048 λ2 @ 20%	2.71s	6.12s	2.26×
BC ω1024 λ2 @ 20%	1.44s	3.13s	2.17×
BC ω1024 λ3 @ 20%	1.82s	4.01s	2.21×
BC ω512 λ2 @ 20%	0.80s	1.67s	2.08×
BC ω1024 λ4 @ 10%	**1.04s**	**1.05s**	**1.01×**

The slowdown tracks FFT-*eligibility* precisely. The one row that is unchanged is the one row whose local length does not divide $p - 1$, so it never moved to roots of unity and never paid for them. The microbenchmark above accounts for essentially all of the gap.

This earlier table is not stored in a data file. The λ=2 and λ=3 rows are re-measured by `decoder_version_benchmark.py` (the three-version table above, 2.15–2.37×, which is the measurement to cite); the λ4 row can be repeated by checking the old file out beside the current one and running the same decodes against each:

```
git worktree add /tmp/prefft 8d93ae3
```

What this means for the parameterisations swept here

Three separate conclusions, which are easy to run together and should not be:

Phase 1's NTT path is correct but not yet worth using. It needs the Newton-iteration division before its asymptotics reach the measured cost. Until then the Vandermonde solver is the faster path, which is why the current decoder's phase 1 uses it alone.

Phase 2's NTT path cannot work as written, and this is not a speed question. It relabels the union of two adjacent supports with a fresh set of n_{aux}-th roots of unity and decodes it as a flat GRS code. The union is not one — as the section above on the pairwise phase already notes, the two blocks' parity checks assign different evaluation points to the columns they share, and relabelling discards exactly the structure that would be needed. This is the same gap identified there: the paper's Algorithm 1 auxiliary GRS construction is the missing piece, and it would fix the correction radius and the cost together.

Moving to roots of unity is a bet, not an improvement. It costs a factor of about 2.3 today (2.33× median across the measured points) and buys nothing until the division is fixed. It is the right bet *if* the Newton iteration lands, because then the same choice that costs 2.3× unlocks an $O(n \log^2 n)$ decoder. It is the wrong one if the work stops here.

Reproducing

```
pip install -r requirements.txt

python3 benchmark_1D_2D_BC.py     # the erasure sweep -> benchmark_results.json
python3 fft_benchmark.py          # the NTT measurements -> fft_results.json
python3 test_bc_fft.py            # structural checks on both decoder phases
python3 make_figures.py           # every figure -> docs/figures/
python3 make_docs.py              # every report -> docs/*.pdf
```

Per-trial results — erasure counts, decode times, success flags, and the decoder's own claim separately from ground truth — are written to `benchmark_results.json`. The NTT comparison, its profile breakdown, the node-width microbenchmark and the phase-by-phase structural check go to `fft_results.json`.

The erasure sweep takes about two hours. `fft_benchmark.py` took 10–20 minutes when it could still run the NTT comparison, because it deliberately runs the slow path it is measuring; with the current decoder that comparison is carried forward rather than repeated (see the NTT section). Neither is a prerequisite for the figures. `make_figures.py` reads the committed JSON, so every figure in this report regenerates in under a minute without re-measuring anything — most of that is the bootstrap behind the threshold intervals.

That indirection is the point. These figures were previously hand-written SVG, which meant a re-run could leave a chart quietly disagreeing with the table printed beside it. Every figure here is now computed from the same file the prose quotes, and the predictions drawn on them — the rate bound $1-R$, the $1/(1+\lambda)$ local reach, the p^4 and hypergeometric curves in the network report — are calculated in the plotting code rather than transcribed into it.